

European Bank Customer Churn Prediction & Risk Scoring

**Business Analyst Research Paper
Prepared Using Python Pandas and Streamlit**

Prepared by: Varadaraj H S

Date : 26 September 2026

Abstract.

Customer churn is a material retention problem for banking organizations because identifying customers at elevated risk enables more targeted intervention. This project develops a customer-churn predictions and risk-scoring workflow using the supplied European Bank Dataset. The notebook analyzes 10,000 observations and 14 original columns, engineers four additional predictors, and evaluates Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and XGBoost models on a stratified 80/20 split. Random Forest is used for risk categorization, explainability, threshold analysis and scenario simulation. On the held-out test data, Random Forest achieved accuracy 0.8215, recall 0.7150, F1-score 0.6198 and ROC-AUC 0.8651. Gradient Boosting had the largest notebook ROC-AUC(0.8692); XGBoost had 0.8686. Risk Score ranged from 0.0224 to 0.9780 and were segmented as low (<0.30), medium($0.30 < 0.60$) and high (≥ 0.60) risk. Results are descriptive of this dataset and split; they do not establish causal effects or production performance.

Keywords: customer churn; banking analytics; Random Forest; Gradient Boosting; XGBoost; SHAP; risk scoring; threshold optimization.

1. Introduction

Customer churn prediction translates customer-level records into estimates of the probability that a customer exits. In this project, the target variable is Exited. The notebook combines exploratory analysis, supervised classification, explainability and scenario testing to support a European Bank risk-scoring use case. It is predictive rather than causal: probabilities can aid prioritization but do not establish that changing an input changes customer behaviour.

2. Problem Statement and Objectives

The notebook defines a solution that predicts churn, generates individual probability scores, identifies key drivers, reduces false-positive predictions, improves interpretability using explainable AI, segments customers by risk and supports retention recommendations. The operational task is therefore

to distinguish customers with different estimated risks while exposing accuracy, precision and recall trade-offs.

3. Literature Review - Evidence Boundary

Placeholder: no external literature or bibliography was supplied with the notebook. This paper therefore does not make literature-derived claims or fabricate a review. A final academic submission should add verified sources on bank churn, class imbalance, calibration and explainable machine learning.

4. Dataset and Data Preparation

The notebook loads `European_Bank.csv` with 10,000 rows and 14 columns. Every column is complete, no duplicate rows were found and removing duplicates retained 10,000 rows. Year has a single value (2025) and CustomerId, Surname, Year and EDA-only AgeGroup were excluded from modeling.

Field Group	Notebook-derived details
Target	Exited (0=retained; 1= churned)
Categorical Input	Geography(3 levels); Gender(2 levels)
Inputs	CreditScore, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary
Class Balance	7,963 retained (79.63%); 2,037 churned (20.37%)

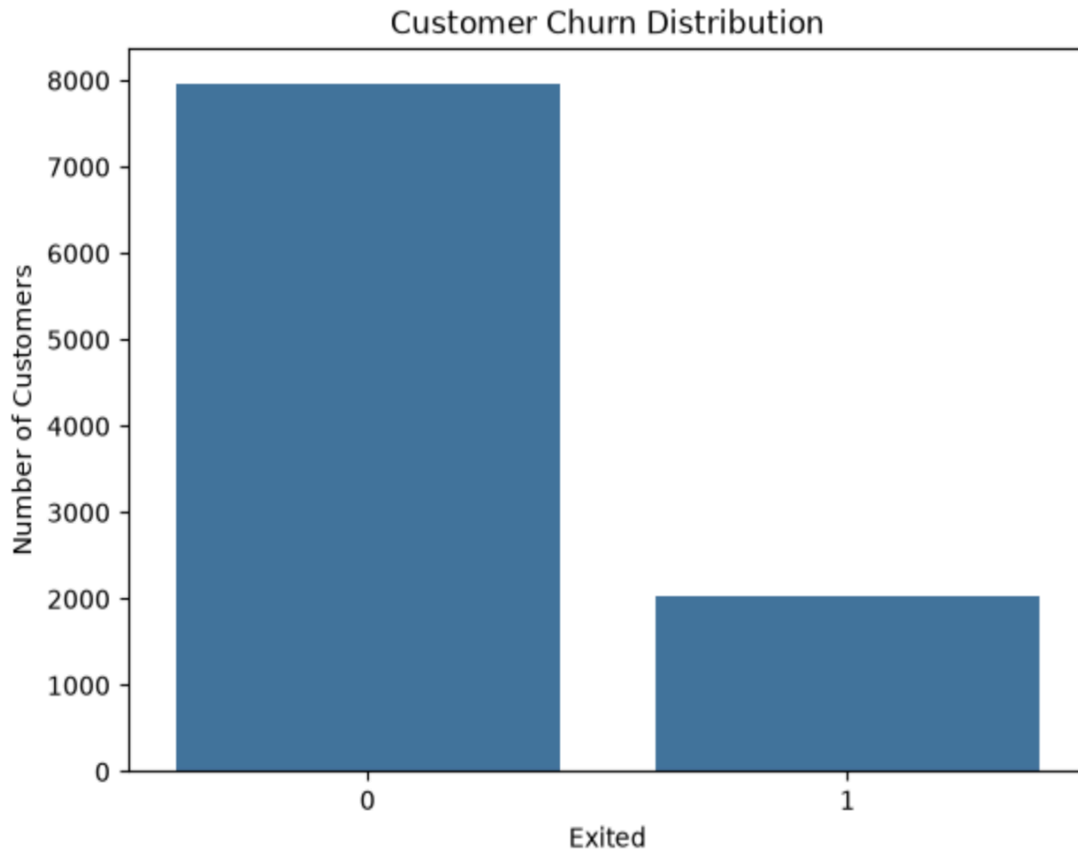


Figure 1. Customer Churn Distribution Recorded in the Notebook

5. Exploratory Data Analysis

The overall churn rate is 20.37%. Germany had the largest geography-level rate (32.44%), compared with France (16.15%) and Spain (16.67%). Female customers had 25.07% churn versus 16.46% for male customers. Inactive customers had 26.85% churn versus 14.27% for active members. These are associations in the dataset, not causal findings.

EDA Insights 1 - Geography: Germany has a churn rate approximately twice that of France and Spain in this dataset. This identifies a segment for deeper service, pricing, product and data-quality investigation; it does not demonstrate that geography itself causes churn.

EDA Insights 2 - Engagement: Inactive members show a 12.58 percentage-point higher observed churn rate than active members (26.85%

versus 14.27%). Engagement status is therefore a useful prioritization signal for retention monitoring.

EDA Insights 3 - Product relationship: Customers with two products have the lowest observed churn (7.58%), while the recorded rates for three and four products are much higher (82.71% and 100%). Because group sizes are not reported, these extreme product-count rates must be validated before any policy is designed around them.

EDA Insights 4 - Age Pattern: Churn rises from 7.52% for customers under 30 to 56.21% for the 50-60 group, then is 24.78% for 60+. Age is also the highest Random Forest importance feature, but the combined evidence remains predictive descriptive rather than causal.

Segment	Churn Rate
France	16.15%
Germany	32.44%
Spain	16.67%
Female	25.07%
Male	16.46%
Inactive Member	26.85%

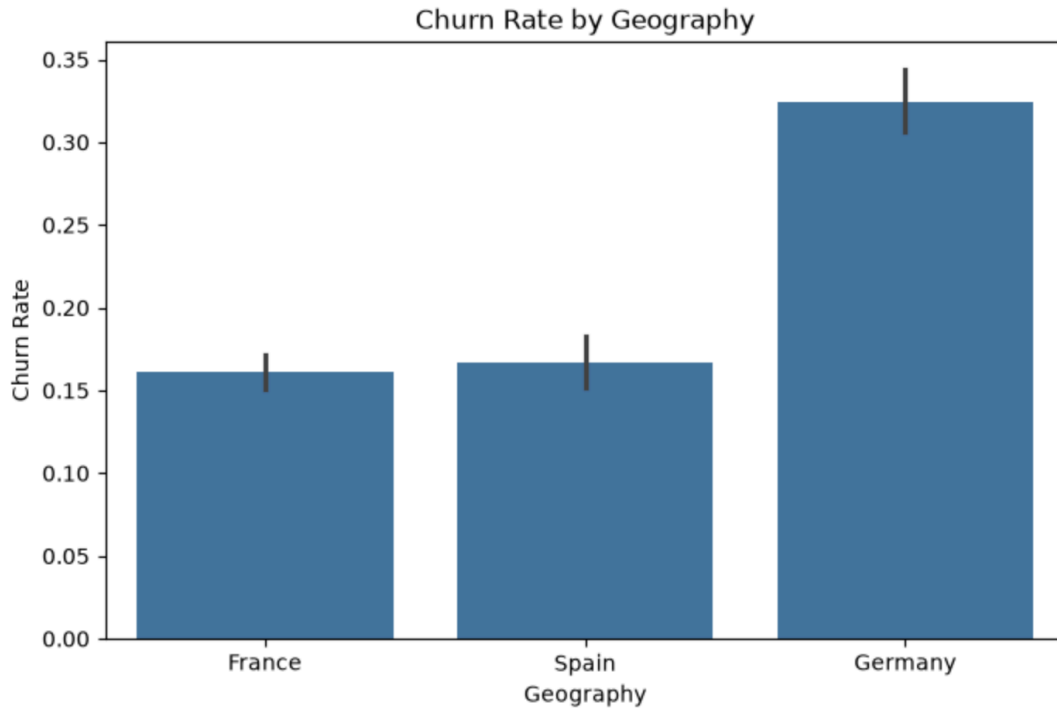


Figure 2. Churn rate by geography

Product-count rates were 27.71% (one), 7.58% (two), 82.71%(three), and 100% (four); the notebook does not give group counts, so the last two rates require caution. Age-group churn rose from 7.52% (<30) to 56.21% (50-60), then was 24.78% for 60+.

6. Feature Engineering and Preprocessing

The notebook creates $\text{BalanceSalaryRatio} = \text{Balance} / (\text{EstimatedSalary} + 1)$, $\text{ProductDensity} = \text{NumOfProducts} / (\text{Tenure} + 1)$, $\text{EngagementProduct} = \text{IsActiveMember} \times \text{NumOfProducts}$, and $\text{AgeTenure} = \text{Age} \times \text{Tenure}$. It uses 12 numerical and two categorical features. `StandardScaler` is applied to numerical features; `OneHotEncoder(drop="first",handle_unknown="ignore")` is applied to categories. A stratified 80/20 split with `random_state=42` created 8,000 training and 2,000 test records.

Engineered Feature	Definition
BalanceSalaryRatio	$\text{Balance} / (\text{EstimatedSalary} + 1)$
ProductDensity	$\text{NumOfProducts} / (\text{Tenure} + 1)$
EngagementProduct	$\text{IsActiveMember} \times \text{NumOfProducts}$

Age Tenure	Age x Tenure
------------	--------------

7. Methodology

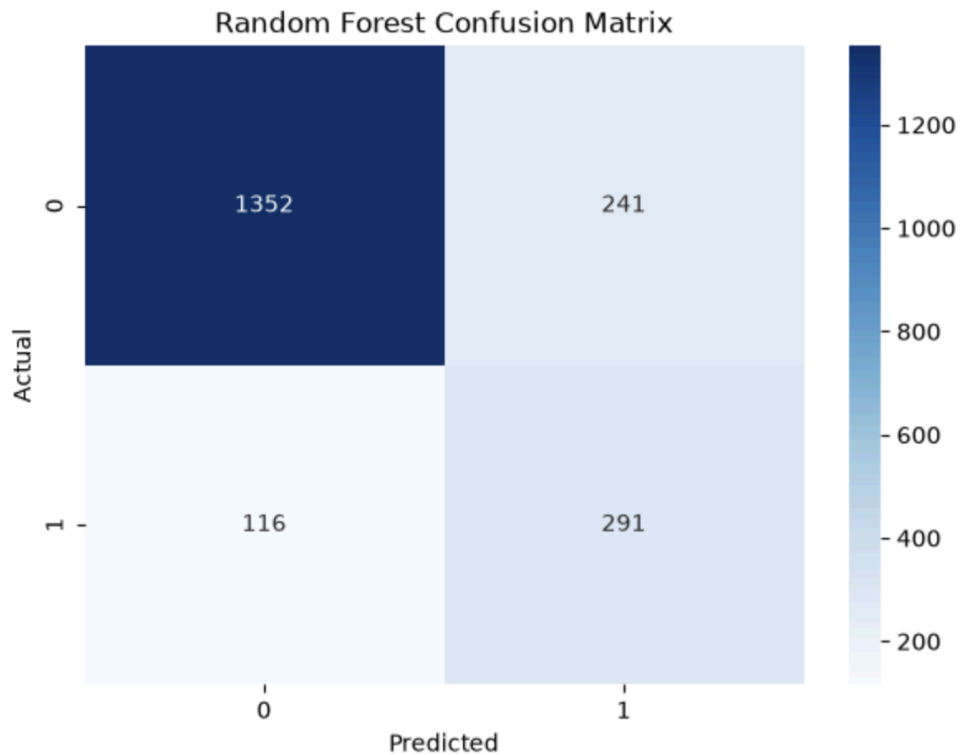
Five common-preprocessing pipelines were fitted. Logistic Regression used max_iter=1000. Decision Tree used max_depth=5. Random Forest used 300 estimators, max_depth=10, random_state=42 and class_weight="balanced". Gradient Boosting used 200 estimators, learning_rate=0.05, max_depth=3. XGBoost used 200 estimators, learning_rate=0.05, max_depth=4 and eval_metric="logloss". Accuracy, precision, recall, F1-score and ROC-AUC were calculated on the test split. Random Forest also underwent 5-fold stratified ROC-AUC cross-validation.

8. Model Evaluation

Gradient Boosting had the top notebook ROC-AUC (0.8692), closely followed by XGBoost (0.8668) and Random Forest (0.8651). Random Forest had the strongest recall (0.7150) and F1-score (0.6198). The notebook uses Random Forest for subsequent risk scoring and explainability; that is a workflow choice, not a claim of universal superiority.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting	0.8695	0.7897	0.4889	0.6039	0.8692
XGBoost	0.8655	0.7717	0.4816	0.5930	0.8668
Random Forest	0.8215	0.5470	0.7150	0.6198	0.8651
Decision Tree	0.8570	0.7951	0.4005	0.5327	0.8442
Logistic Regression	0.8060	0.5704	0.1892	0.2841	0.7735

Random Forest 5-fold ROC-AUC scores were[0.8620, 0.8468, 0.8573, 0.8626, 0.8728], mean 0.8603 (SD 0.0084)



9. Churn Risk Scoring

Random Forest probabilities ranged from 0.0224 to 0.9780. The notebook assigns Low Risk(<0.30), Medium Risk (0.30 to <0.60) and High Risk (≥ 0.60). Of 2,000 test customers, 934(46.7%) were low risk, 707 (35.4%) medium risk and 359 (18%) high risk. The binary predictions cutoff is 0.50.

Risk Category	Probability Rule	Customers	Share
Low	<0.30	934	46.7%
Medium	0.30 to <0.60	707359	35.4%
High	≥ 0.60		18%

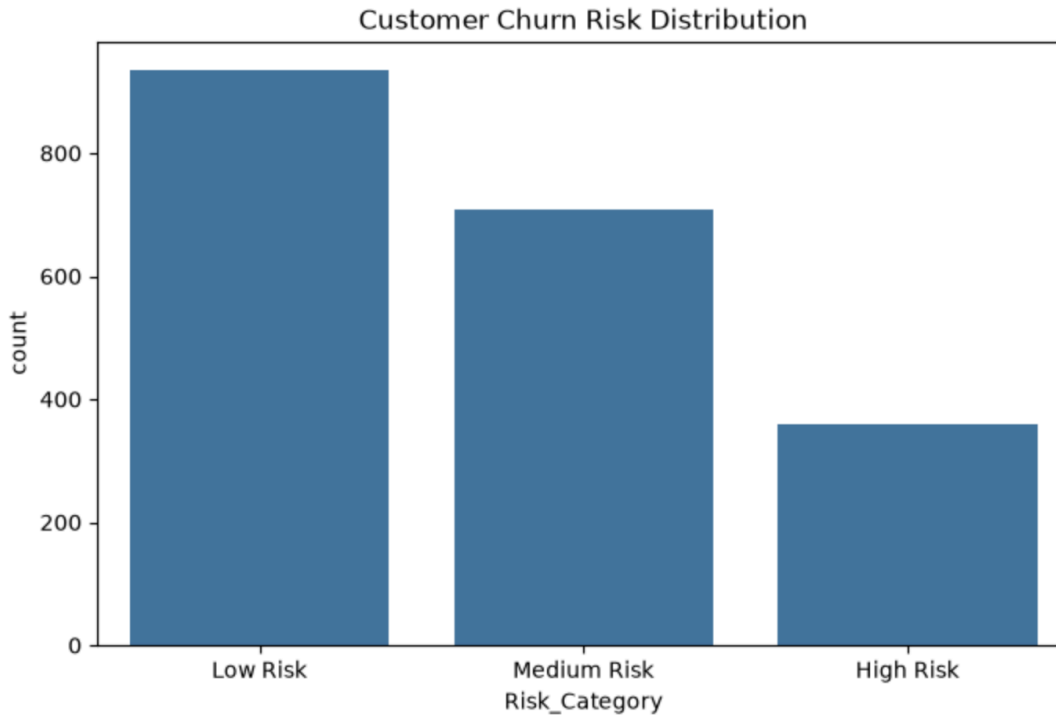


Figure 4: Notebook Distribution of Customer Risk Categories

10. Threshold Analysis and False-Positive Trade offs

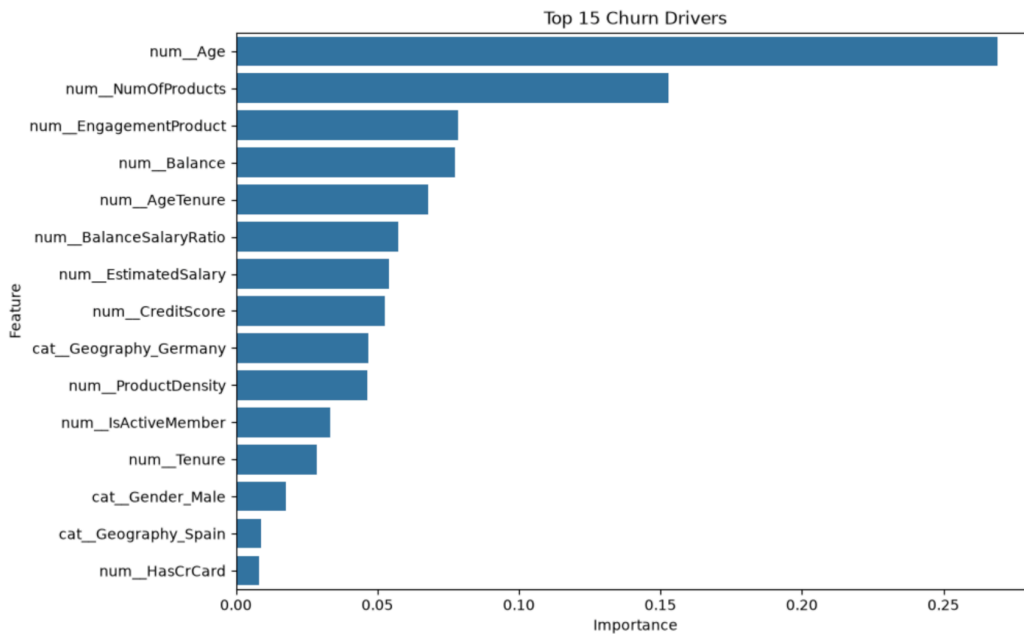
The threshold grid shows its highest F1-score at 0.50 (0.6198), with 241 false positives and 116 false negatives. At 0.7, false positives fall to 61 and precision rises to 0.7672, but recall declines to 0.4939 and false negatives increase to 206. Threshold selection is a business trade-off: contact cost, retention value and determine the suitable policy.

Threshold	Precision	Recall	F1	False Positives	False Negatives
0.30	0.3471	0.9091	0.5024	696	37
0.40	0.4364	0.8010	0.5650	421	81
0.50	0.5470	0.7150	0.6198	241	116
0.60	0.6602	0.5823	0.6188	122	170
0.70	0.7672	0.4939	0.6009	61	206

11. Explainability

Random Forest feature importance ranks Age (0.2690) and NumOfProducts (0.1530) first, followed by EngagementProduct (0.0773), Balance (0.0773) and AgeTenure (0.0679). The notebook also runs SHAP TreeExplainer, creates a global summary, a local waterfall plot for test customer index 0, and partial-dependence displays. These are model explanations, not causal effects.

Rank	Feature	Importance
1	Age	0.2690
2	NumOfProducts	0.1530
3	EngagementProduct	0.0784
4	Balance	0.0773
5	Age Tenure	0.0679
6	BalanceSalaryRatio	0.0575
7	EstimatedSalary	0.0541
8	CreditScore	0.0527



12. What-If Scenario Analysis

For one held-out customer (test index 0), the current model probability is 8.94%. Setting IsActiveMember to 1 yields 11.38%, a +2.43 percentage-point change. Setting NumOfProducts to 2 yields 8.94%, a 0.00-point change. These are predictive counterfactual outputs for a single record, not proof that these changes cause customer behaviour; The notebook does not report the full original customer profile in the summary.

Scenario	Churn Probability	Change
Current Customers	8.94%	-
Active member Scenario	11.38%	+2.43 points
2 Products	8.94%	0.00 points

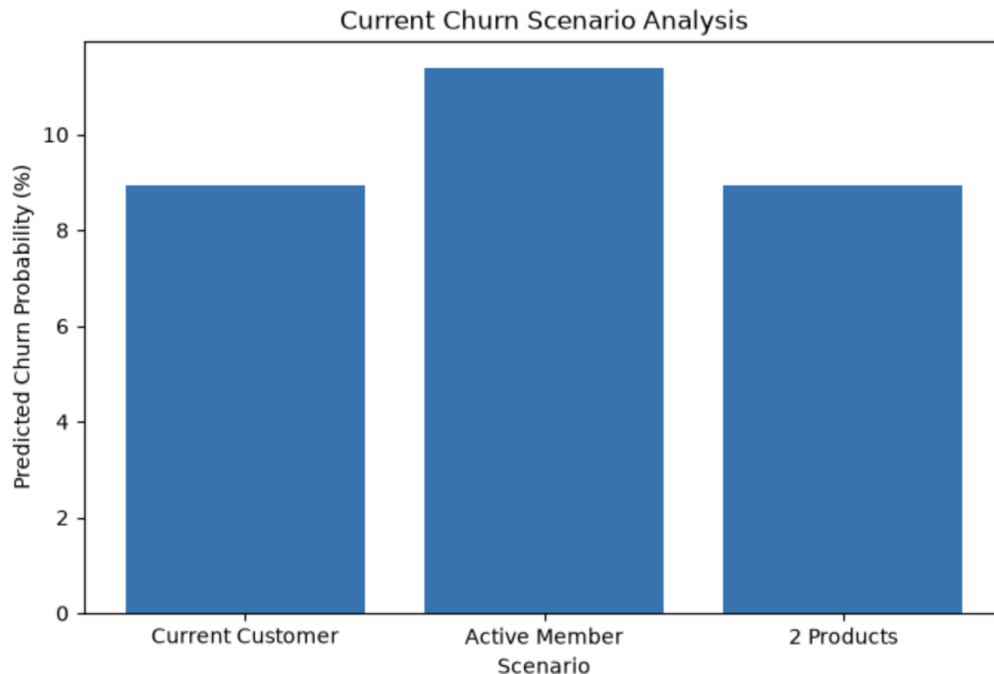


Figure 7 What-if Scenario comparison

13. Streamlit Deployment Outputs

Describe the Streamlit application as the practical deployment of the predictive model. Explain that the interface provides an interactive customer risk calculator, churn probability, binary churn prediction, risk category, decision threshold, probability distribution, feature importance, what-if scenario analysis, scenario comparison and customer risk-category visualization.

14. Results and Discussion

Discuss the model evaluation together with the deployed application's output. Explain the distinction between a continuous churn probability and a threshold-based churn classification. The displayed 23.5% prediction is below the 30% low-risk boundary and below the 50% binary decision threshold. Discuss the probability-distribution visualization, feature-importance dashboard, and what-if simulator. The displayed what-if scenario produces the same 23.5% churn probability as the original customer, resulting in a 0 percentage-point change. Make clear that this is a model based scenario simulation and not evidence of a causal relationship. The applications also reports 10,000 total customers 2,037

churned customers and an overall churn rate of approximately 20.4% in its dataset-information panel. These application outputs complement the model evaluation and demonstrate how the predictive analysis can be presented as an interactive decision-support tool.

15. Business Recommendations

1. **Use risk scores as a retention-prioritization tool.** Use Low, Medium and high risk categories and the underlying churn probability to organize customer-review queues, while retaining appropriate human oversight.
2. **Select the operating threshold using business costs.** It demonstrates a trade-off between false positive and false negative. Threshold selection should consider contact cost, retention value, campaign capacity and the cost of missed churners.
3. **Investigate engagement as a retention signal.** Use the observed engagement patterns and the what-if simulator to identify customers for further investigation. Validate proposed interventions experimentally before treating them as causal solutions.
4. **Investigate geography and customer segments.** Differences across geographic groups can guide further investigation of service experience, product mix, tenure, customer journeys and data quality rather than being treated as causal explanations.
5. **Validate unusual product-count patterns.** Verify group sizes, product definitions and data quality before designing product-specific retention policies.
6. **Use explainability to support analyst review.** Feature importance, SHAP and partial-dependence outputs can help analysts understand model behaviour. They should investigate rather than be interpreted as causal proof.
7. **Monitor the model after deployment.** Track realized churn, false positives, false negatives, drift and subgroup performance and reassess thresholds as business conditions change.

16. Limitations

The datasets representativeness of the wider banking customer population is not established. Therefore results should not automatically be generalized to other banks, regions or time periods. Most model comparisons are based on a train/test split, so performance may vary under different samples or future data. Temporal and external validation were not established in the project. The analysis does not establish probability calibration or a formal fairness audit across customer groups. These would be important before operational deployment. Some product-count groups show unusual churn patterns, but the corresponding group sizes should be checked before drawing strong conclusions. The what-if simulator provides a model-based counterfactual scenario. It does not demonstrate that changing a customer attribution would causally change actual churn behaviour.

17. Future Work

Future research should evaluate the model using temporal validations and where possible, an independent external dataset. This would provide stronger evidence of generalization. Probability calibration and cost-sensitive threshold optimization could be added so that risk scores and operational decisions better reflect the bank's retention objectives. Future versions can include formal fairness and subgroups-performance analysis, border hyperparameter tuning, model monitoring, drift detection, model versioning and governance controls. Retention strategies suggested by the model should be evaluated through controlled experiments or other appropriate causal designs rather than relying solely on predictive associations.

18. Conclusion

The European Bank customer churn project presents an end-to-end predictive analytics workflow covering exploratory analysis, preprocessing, feature engineering, model comparison, risk scoring, explainability, scenario analysis and interactive Streamlit deployment. The deployed application converts model predictions into an accessible customer-level

risk assessment. It combines churn probability, risk categories, feature importance, probability distribution and what-if analysis to support practical retention analysis. The project demonstrates how machine learning can be integrated into a business analytics workflow while also highlighting the importance of validation, threshold selection, explainability, monitoring and responsible interpretation before operational use.

Appendix : Streamlit Application Output Screenshots

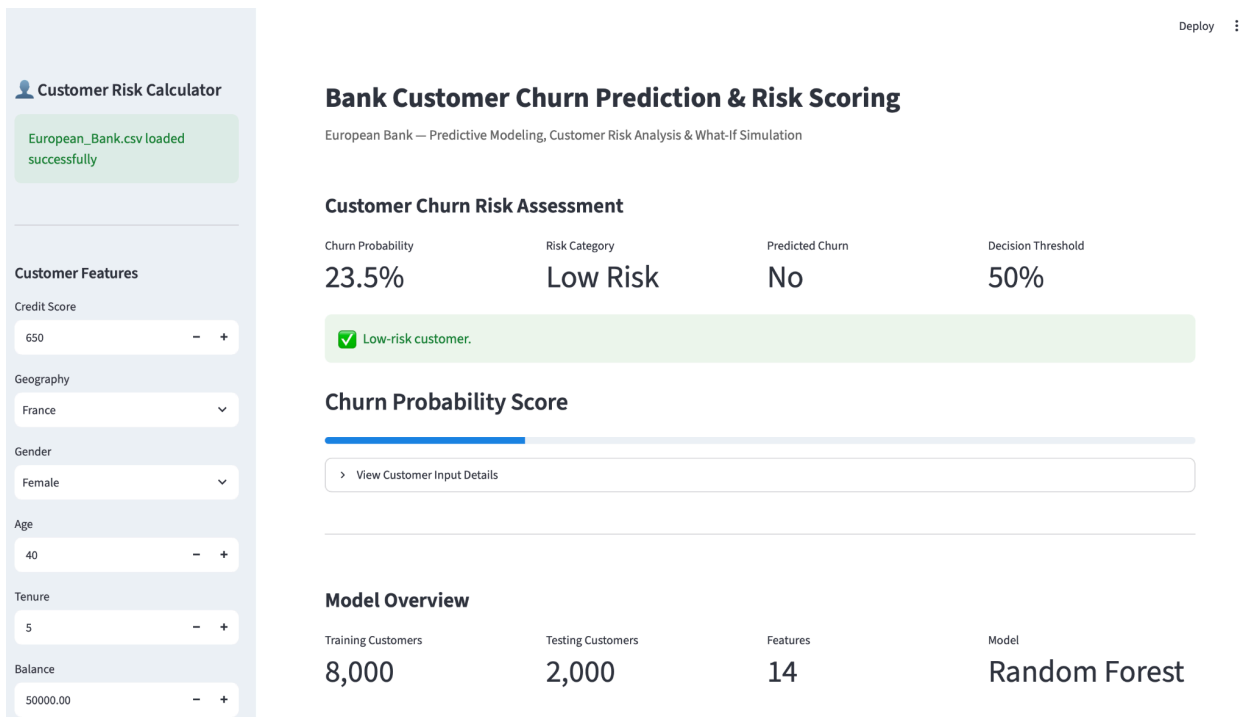


Figure 1: Bank Customer Churn Predictions & Risk Scoring

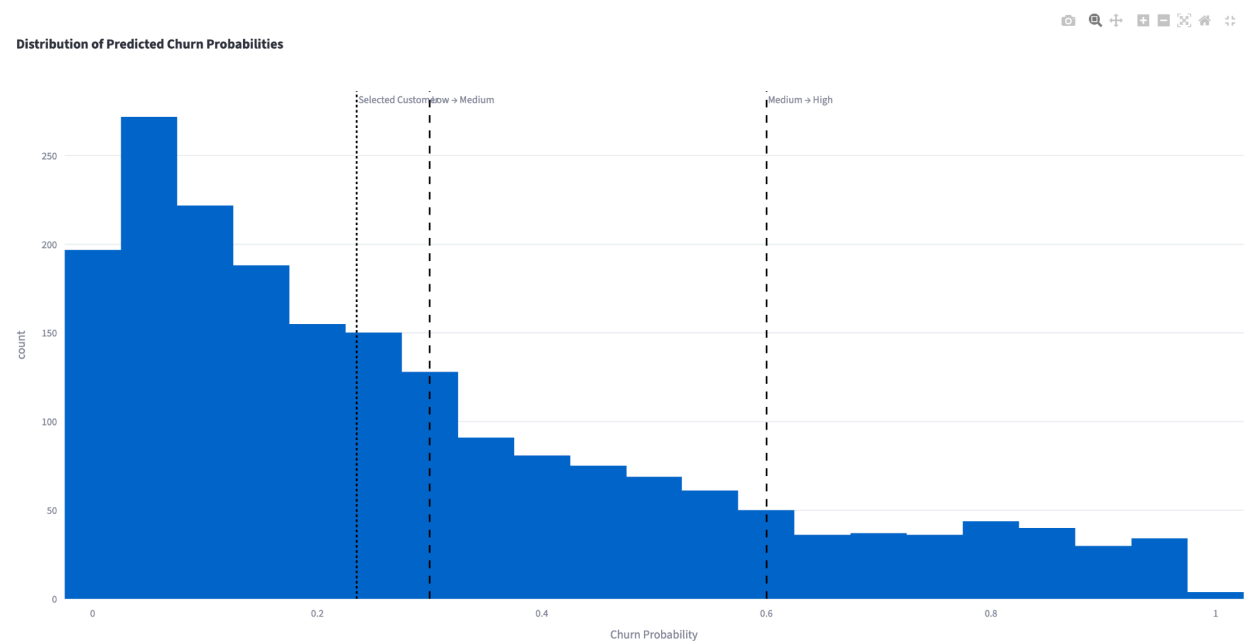


Figure 2: Distribution of Predicted Churn Probabilities

Top 15 Features Influencing Customer Churn

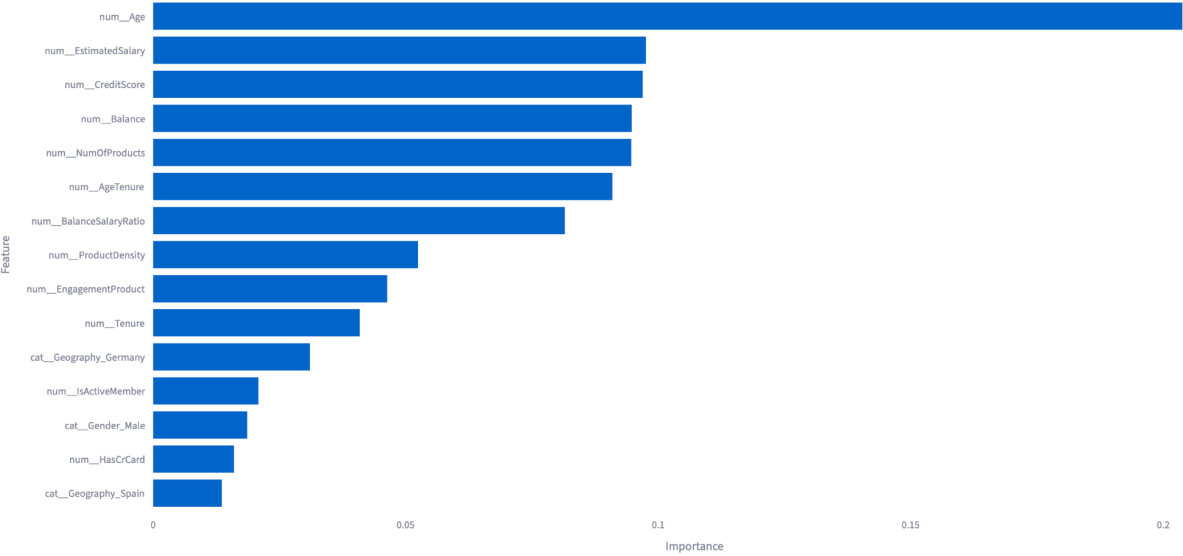


Figure 3: Top 15 Feature Influencing Customer Churn

geography

France

Gender

Female

Age

40

Tenure

5

Balance

50000.00

Number of Products

2

Has Credit Card?

No

Active Member?

No

Estimated Salary

50000.00

Deploy

Feature	Importance
num__Age	0.2038
num__EstimatedSalary	0.0976
num__CreditScore	0.0969
num__Balance	0.0948
num__NumOfProducts	0.0947
num__AgeTenure	0.0909
num__BalanceSalaryRatio	0.0815
num__ProductDensity	0.0525
num__EngagementProduct	0.0463
num__Tenure	0.0409

What-If Scenario Simulator

Change customer engagement and number of products to observe how the predicted churn probability changes.

Scenario — Active Member

Inactive

Scenario — Number of Products

2

What-If Scenario Simulator

Change customer engagement and number of products to observe how the predicted churn probability changes.

Scenario — Active Member

Inactive

Scenario — Number of Products

2

Original Churn Risk

23.5%

Scenario Churn Risk

23.5%

Probability Change

+0.0%

The selected scenario does not change the predicted churn probability.

Scenario Comparison

Scenario	Active Member	Number of Products	Churn Probability
Original Customer	No	2	23.50%
What-If Scenario	No	2	23.50%

Figure 5: What-If Scenario Simulator

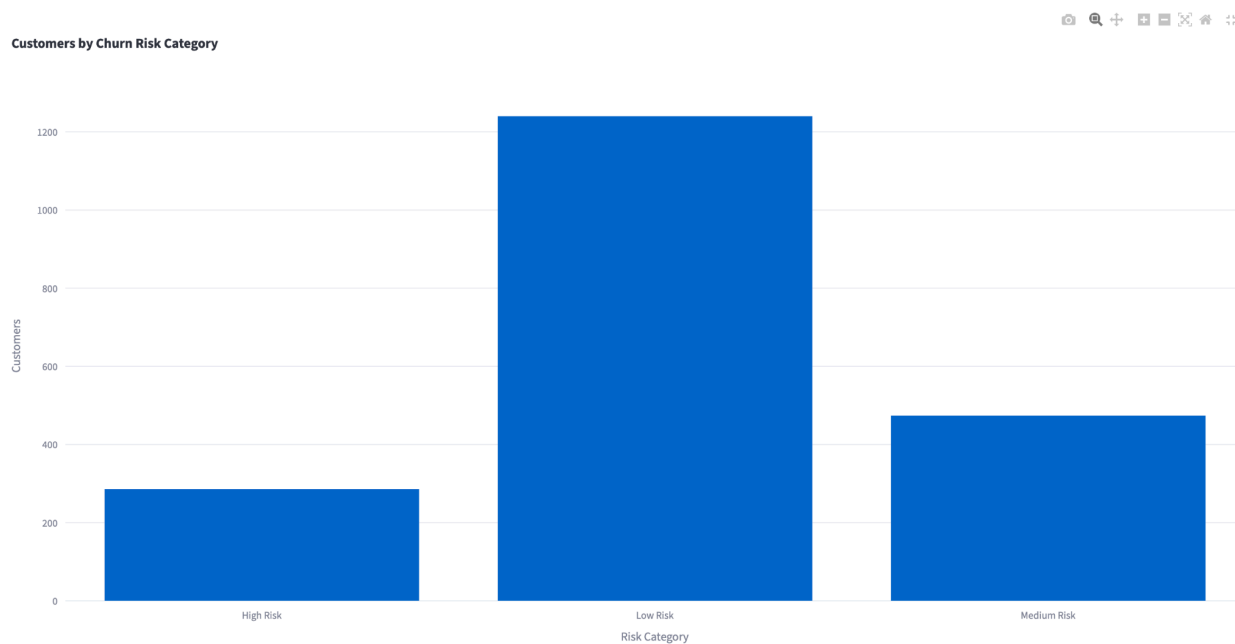


Figure 6 : Customers by Churn Risk Category